

Set answer 2

Scenario based answer

1. Word Cloud

- Collect the product reviews.
- Clean and preprocess the text.
- Remove stop words and unnecessary characters.
- Count the frequency of words.
- Generate the word cloud using word frequencies.
- Display the most frequent words visually.

2. Sentiment Analysis

- Collect the hotel reviews.
- Clean and preprocess the text.
- Convert the text into a suitable representation.
- Apply a sentiment analysis model.
- Classify each review based on sentiment.
- Analyze the overall customer sentiment.

3. Topic Modeling

- Collect the unlabeled news articles.
- Clean and preprocess the text.
- Convert the documents into numerical representations.
- Apply a topic modeling algorithm.
- Identify the important words associated with each topic.
- Assign documents to their relevant topics.

4. CountVectorizer

- Collect the customer messages.
- Clean and preprocess the text.
- Create a vocabulary from the documents.
- Count the occurrence of each vocabulary word in each document.
- Convert the text into a numerical matrix.
- Use this matrix as input for the machine learning model.

5. CountVectorizer vs TF-IDF

- Understand the purpose of the text analysis.
- CountVectorizer represents words based on their frequency.
- TF-IDF considers both word frequency and how common the word is across documents.
- For identifying important terms in job descriptions, TF-IDF can reduce the importance of words appearing across many documents.
- Select the vectorizer based on whether simple word frequency or term importance is required.

6. TF-IDF for Resume

- Collect the resume text.
- Clean and preprocess the text.
- Calculate Term Frequency (TF).
- Calculate Inverse Document Frequency (IDF).
- Combine TF and IDF to obtain a score for each term.
- Rank terms based on their TF-IDF scores.
- Identify terms that are relatively unique and relevant.